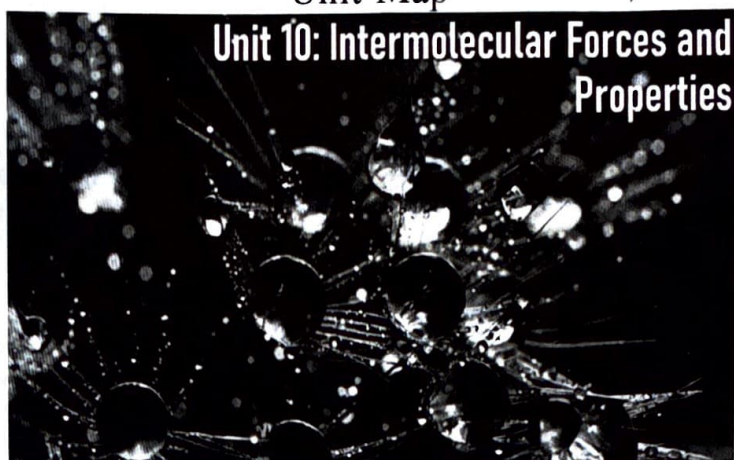


Name: _____ Period: _____

Unit 10: Intermolecular Forces and Properties

AP Classroom Topic 3, Partial 7

Unit Map



Learning Targets & Success Criteria

Textbook Reference: Zumdahl, 5, 10, 11, 16

Date	Day	Learning Target	Activities	Work on this
3/25-26	U10D1	How do gases behave?	Wrap up bonding practice Begin U10 IMF notes	REVIEW FOR AP EXAM
3/27-30	U10D2	What are intermolecular forces?		U10 Textbook Probs #1 REVIEW FOR AP EXAM
3/31-4/1	U10D3	How do IMFs help predict properties?	IMF Card Sort	U10 Textbook Probs #1 REVIEW FOR AP EXAM
4/2-6	U10D4	How do IMFs determine phase and solubility?	Lab: Calming Bottle	U10 Textbook Probs #1 REVIEW FOR AP EXAM
4/7-8	U10D5	What determines precipitation?	~IMFs Quiz	U10 Textbook Probs #2 REVIEW FOR AP EXAM
4/9-10	U10D6	How does solubility relate to equilibrium? What is the common ion effect?		U10 Textbook Probs #2 REVIEW FOR AP EXAM
4/13-14	U10D7	What have we learned about aqueous equilibria?	Ksp Quiz	REVIEW FOR AP EXAM

Ideal Gases: Not Attractive at all

Kinetic Molecular Theory of Gases

- a simple model that attempts to explain properties of an ideal gas
- based on an assumption about the particles: that they experience NO intermolecular attraction at all

Ideal gases consist of particles that have the following properties:

- The particles are so small compared to the distances between them that the volume of the individual particles can be assumed to be negligible (zero), but their mass is NOT negligible.
- The particles are in constant motion. The force of the collisions of the particles with the walls of the container are the cause of the pressure exerted by the gas.
- The particles are assumed to exert no forces on each other; they are assumed neither to attract nor to repel each other.
- The average molecular kinetic energy of a collection of gas particles is assumed to be directly proportional to the Kelvin temp of the gas.

-Because ideal gases conform to these assumptions, we can describe its P, V, n, and T with simple mathematical models, the Ideal Gas Law:

$$PV = nRT$$

-BUT real gases don't conform to these assumptions!!!

Application: Molar Mass from Gases

- Recall that the Ideal Gas Law, $PV = nRT$, is useful for determining moles of a gas for stoichiometry and other uses
- Also, when paired with mass data, the molar mass of a gas can be determined:

$$MM = \text{mass}/\text{moles} = m/n$$

Practice: A sample of a certain gas occupies 8.5 L at 1°C and 0.961 atm and has a mass of 6.1 grams. What is the molar mass of the gas?

$$MM = \frac{m}{n} = \frac{6.1\text{ g}}{n} = \frac{6.1}{0.3633} = 17\text{ g/mol}$$

$$PV = nRT \quad 0.961 \cdot 8.5 = n \cdot 0.08206 \cdot 274$$

$$n = 0.3633\text{ mol}$$

Mystery Gas! In the basement of the old chemistry building at Mystery University, a gas cylinder has been found. The gas needs to be disposed of properly, but in order to do that, the identity of the gas needs to be determined. The sealed 7.5L gas cylinder is held at 299K. The pressure gauge reads ~~26500 Torr~~ 34.9 atm . The mass in the cylinder is 404 g and through other chemical tests, it is known to be a halogen, what halogen is it?

$$MM = \frac{m}{n} = \frac{404}{10.67} = 37.87 \sim 38$$

$$PV = nRT$$

$$34.9 \times 7.5 = n \cdot 0.08206 \cdot 299$$

$$n = 10.67$$



Understanding Gas Pressure

This is a schematic diagram showing gas molecules in a container. The molecules are constantly moving in random directions. When a molecule hits the container wall (green), it exerts a tiny force on the wall. The sum of these tiny forces, divided by the interior surface area of the container, is the pressure. (Longer arrows indicate faster velocities, shorter arrows indicate slower velocities)



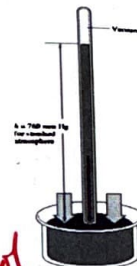
Why aren't all the molecules moving with the same velocity and how is this related to temperature?

average molecular KE

If the temperature increases, what happens to the velocities? To the pressure?

increase, increase

Barometer- measures pressure in mm Hg



MCQ Practice:

1. Two flexible containers for gases are at the same temperature and pressure. One holds 0.50 gram of hydrogen and the other holds 8.0 grams of oxygen. Which of the following statements regarding these gas samples is FALSE? → 0.25 mol

a. The volume of the H_2 container is the same as the volume of the O_2 container. T (same moles)

b. The number of molecules in the H_2 container is the same as the number of molecules in the O_2 container. T (same moles)

c. The average kinetic energy of the H_2 molecules is the same as the average kinetic energy of the O_2 molecules. (T same T)

d. The average speed of the H_2 molecules is the same as the average speed of the O_2 molecules. F - smaller particles move faster

2. A sample of an ideal gas is cooled from $50.0^\circ C$ to $25.0^\circ C$ in a sealed container of constant volume. Which of the following values for the gas will decrease? T ↓

a. The average molecular mass of the gas

b. The average distance between the molecules

c. The average speed of the molecules

d. The average distance between the molecules and the average speed of the molecules

Molar Volume

Molar Volume: At standard temperature and pressure, 1 mol of an ideal gas = 22.4 L

STP = 0 $^\circ C$ and 1 atm

DANGER! DANGER! DANGER! In AP Chem FRQs, gases are almost NEVER at STP, so you should almost NEVER use this 22.4! It is more likely in MCQs.

Real (nonideal) Gases

- Ideal gas behavior assumes no attractions or repulsions between particles. No gas exactly follows the ideal gas law.
- The stronger the IMFs (attractions) in a substance, the less ideal a gas behaves.
small, non-polar particles are the most ideal gases.
- Also, real gas exhibits behavior closest to ideal behavior at low pressure (widely spaced particles) and high temp (highly energetic molecules). Students also behave most ideally under these conditions. (Summer vacation!)

- If the pressure is too high, the particles are not separated by an infinite distance and can interact, slowing/changing their trajectory.
- If the temperature is too low, the particles slow their speed and can interact, slowing/changing their trajectory.

Practice: Which of the following gases deviates most from ideal behavior?

- a. SO₂ c. N₂
b. CH₄ d. H₂

Effusion and Diffusion

Diffusion- mixing of gases- happens because gases move randomly

Effusion- the passage of a gas through a tiny orifice into an evacuated chamber- happens because of differences in pressure

Rates are determined by mass - lighter gases effuse & diffuse faster than heavier gases.

This affects mixtures of gases

Practice: Equal number of moles of He(g), Ar(g), and Ne(g) are placed in a glass vessel at room temperature. If the vessel has a pinhole-sized leak, which of the following will be true regarding the relative values of the partial pressures of the gases remaining in the vessel after some of the gas mixture has effused?

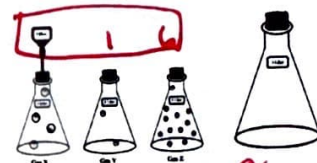
- a. P_{He} < P_{Ne} < P_{Ar} c. P_{Ne} < P_{Ar} < P_{He}
b. P_{He} < P_{Ar} < P_{Ne} d. P_{Ar} < P_{He} < P_{Ne}

Dalton's Law of Partial Pressures

For a mixture of gases in a container, the total pressure exerted is the sum of the pressures that each gas would exert if it were alone.

$$P_{\text{tot}} = P_1 + P_2 + P_3 + \dots$$

Practice: Each one liter flask pictured above is sealed and contains a different noble gas at 298 K. The pressure of Gas X in the first flask is 2.0 atm. The ratio of the numbers of X, Y and Z atoms in the flasks is 2:1:6, respectively. A student then combines all the gases into a previously evacuated 3.0 L flask. What is the total pressure of the gases at 298 K? Hint- read this problem carefully. A. 3.0 atm B. 4.5 atm C. 9.0 atm D. 18 atm

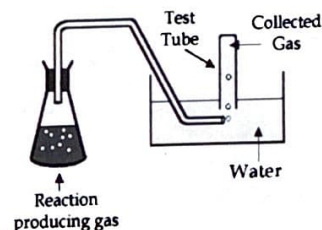


$$\frac{9}{3V} = 3$$

Application: Gases Collected over Water

When gases are collected over water, we must adjust for the pressure of the water vapor.

$$P_{\text{H}_2\text{O}} + P_{\text{gas}} = P_{\text{total}}$$



EXAMPLE 5.11 Collecting Gases over Water

In order to determine the rate of photosynthesis, the oxygen gas emitted by an aquatic plant is collected over water at a temperature of 293 K and a total pressure of 755.2 mmHg. Over a specific time period, a total of 1.02 L of gas is collected. What mass of oxygen gas (in grams) is formed?

$$\begin{aligned} P_T &= P_{\text{H}_2\text{O}} + P_{\text{O}_2} \\ 755.2 &= 17.50 + P_{\text{O}_2} \\ P_{\text{O}_2} &= 737.7 \text{ mmHg} \end{aligned}$$

$$\begin{aligned} PV &= nRT \\ 737.7 \cdot 1.02 &= n \cdot 62.4 \cdot 293 \\ n &= 0.04155 \text{ mol} \\ &= \frac{0.04155 \text{ mol}}{1 \text{ mol}} \cdot 32.00 \text{ g} \\ &= 1.3 \text{ g O}_2 \end{aligned}$$

Intro to IMFs: Sooooo Attractive!

Intramolecular forces: the attraction that results when electrons are given, taken, or shared to form internal bond.

- Ionic bonds
- Covalent bonds
- Metallic bonds

Intermolecular forces (IMFs): the attraction between two or more distinct molecules or particles.

- In a physical change, only intermolecular forces are broken because no chemical bonds are broken (same substance).
- In a chemical change, intramolecular forces are broken as chemical bonds are broken/formed (new substance).

Intramolecular forces are much, MUCH stronger than intermolecular forces!
Think of the children!

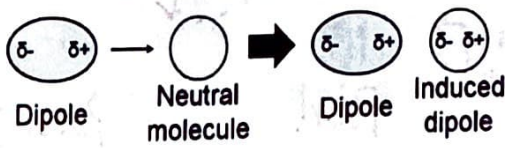
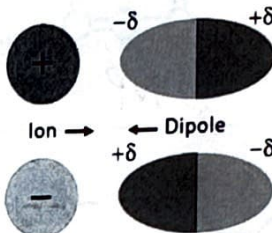


Types of IMFs

Type	Present in	Molecular perspective
London Dispersion Forces/ Van der Waals Forces (LDFs)	• <u>ALL</u> molecules and atoms (nonpolar and polar) • Strength depends on total <u>#</u> of electrons in an atom/molecule	instantaneous dipoles
Dipole-dipole attractive forces	• Between <u>polar</u> molecules (which already have a dipole moment)	
Hydrogen "bonding" attractive forces	• Between H bonded to F, O, or N on one molecule and <u>lone pair</u> on N, O, or F of another molecule → they are electronegative e <u>Not</u> ! → Hydrogen "bonds" are <u>Not</u> actual bonds (intramolecular forces); the name is very misleading!	

If comparing similar-sized particles: **Hydrogen bonding IMFs > dipole-dipole IMFs > LDFs**

Mixed IMFs! Attractive Forces Between Two _____ Compounds

Type	Present in	Molecular perspective
Dipole – induced dipole attractive forces	<u>Polar molecules</u> combined with <u>Nonpolar molecules</u>	
Ion – dipole attractive forces	<u>Ions/ Ionic Compounds</u> combined with Polar molecules	

If comparing **similar-sized** particles:

Ion – Dipole > Hydrogen bonding > Dipole – dipole > Dipole – Induced Dipole > LDFs

In Summary

	Non-polar, single elements	polar + non-polar	polar no H – FON	polar with H – NOF	polar + ions
London dispersion forces (LDFs or Van der Waals)	✓	✓	✓	✓	✓
Dipole – Induced Dipole (multiple compounds)	✗	✓	✗	✗	✗
Dipole – Dipole	✗	✗	✓	✓	✗
Hydrogen “bonding” attractive forces	✗	✗	✗	✓	✗
Ion – Dipole (multiple compounds)	✗	✗	✗	✗	✓

Let's Practice!

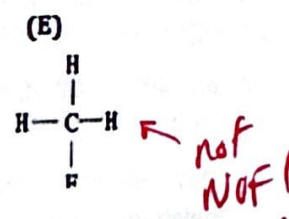
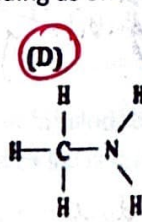
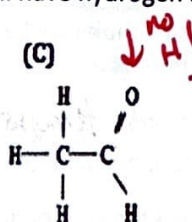
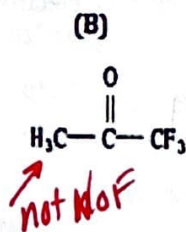
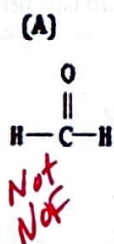
Molecule	IMFs present?
1. Label each as polar, non-polar, or an ion 2. If polar, is it NOF?	1. Which IMFs are present between the compounds? 2. Underline the dominant (strongest) IMF exhibited
CH ₃ CH ₂ CH ₃ NP!	<u>LDF</u>
H—F: P! HNOF!	LDF, dipole-dipole, <u>H bond</u>

no Hbond → CH_3F Polar! + H_2O Polar! NOF!	LDF, dipole dipole, <u>H bond</u>
$\text{O}=\text{C}=\text{O}$ NP	<u>LDF</u>
Guanine + Cytosine P, NOF P, NOF	LDF dipole dipole <u>Hbond</u>
$\text{CH}_3\text{CH}_2\text{OH}$ P, NOF + $\text{O}=\text{C}=\text{O}$ NP	LDF <u>dipole induced dipole</u>

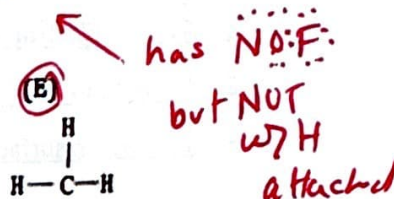
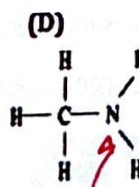
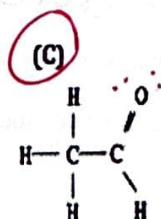
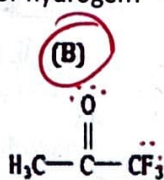
Multiple Choice Practice

- The forces of attraction between the permanent partially positive regions of one molecule and permanent partially negative regions of another nearby molecule are called _____.
 - covalent bonds
 - hydrogen bonding
 - London dispersion forces
 - dipole-dipole forces
- A polar molecule is one in which:
 - ions exist
 - only London dispersion forces exist
 - it has a VSEPR structure that is symmetrical
 - electron density is unequally distributed
- Which of the following processes involves breaking intermolecular forces? (NOT bonds!)
 - $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2 \text{HCl}(\text{g})$
 - $2 \text{C}_2\text{H}_6(\text{g}) + 7 \text{O}_2(\text{g}) \rightarrow 4 \text{CO}_2(\text{g}) + 6 \text{H}_2\text{O}(\text{g})$
 - $\text{I}_2(\text{g}) \rightarrow 2 \text{I}(\text{g})$
 - $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{l})$
- What is the dominant intermolecular force in CH_3OH ? **polar, NOF!**
 - London dispersion forces
 - Ion-dipole attraction
 - Dipole-dipole attraction
 - Hydrogen bonding

5. Which one of the following substances will have hydrogen bonding as one of its intermolecular forces?



6. Circle ALL of the following substances which cannot hydrogen bond with another molecule of the same substance, but would be capable of hydrogen bonding with a different molecule that has H directly bonded to fluorine, oxygen, or hydrogen?



The Language of IMFs (i.e. How to FR)

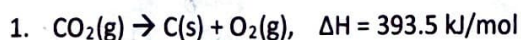
Notes about Language: Talking about IMFs can be tricky! And of course, IMF-based questions are VERY common on free response portion of the AP test, so it's essential to carefully choose your language to avoid losing points.

Most IMF points are lost when student inadvertently imply (or, worse, directly state) that during state changes:

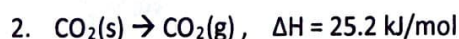
1. Molecular bonds are being broken during state changes (NOPE!)
2. Molecules are being broken apart (NOPE!)

How to Talk about IMFs	How to Talk about Bonds
• <u>overcoming</u> IMFs ○ Avoid breaking IMFs • <u>separating</u> molecules ○ NEVER break apart molecules • IMFs are <u>between</u> molecules	• <u>breaking</u> bonds • Bonds are <u>inside</u> molecules

Free Response Practice: What type of intermolecular OR intramolecular attractive forces must be overcome for each of the following processes to occur?



Covalent bonding

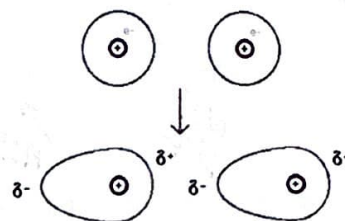


LD F

A Closer Look at London Dispersion Forces (LDFs): Induced Dipole – Induced Dipole Attraction

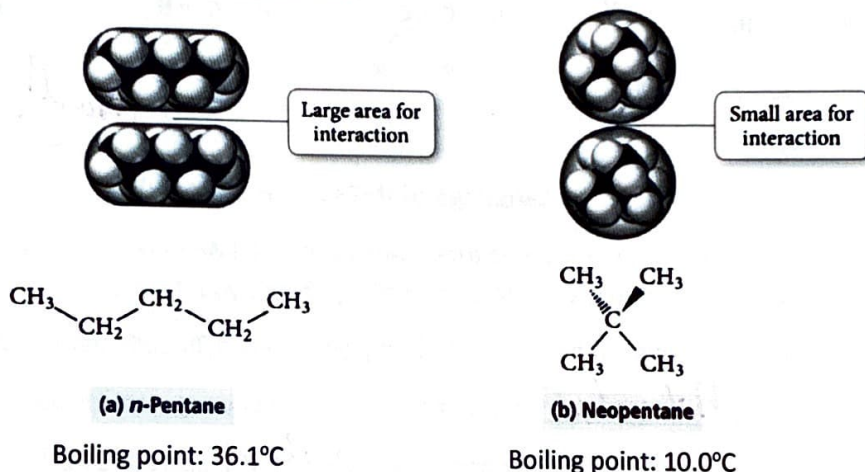
LDFs are determined by the polarizability of a molecule (i.e. how much the electron cloud can temporarily be shifted)

- an electron cloud, even in a nonpolar molecule, can temporarily shift, causing one side of the molecule to be more negative than another
- this temporary dipole can induce (cause) a temporary dipole on a neighboring molecule (hence the name "induced dipole")



What can increase LDFs?

1. Greater electron cloud (more electrons): molecule is more polarizable = ↑ LDFs
2. Increase in molar mass (implies more electrons): molecule is more polarizable = ↑ LDFs
3. Increase in surface-to-surface contact area: ↑ induced dipole = ↑ LDFs



Be careful!

- When non-polar substances with only London dispersion forces have a considerably larger (and thus very polarizable) electron cloud than the polar molecules, the LDFs can be quite substantial and be **STRONGER** than hydrogen bonding forces or dipole-dipole forces (!!)

Example: Cl₂ has a higher boiling point than HCl. Explain.

Even though HCl is a polar molecule & experiences dipole-dipole IMFs, Cl₂ is a large ~~non~~ nonpolar molecule with a polarizable e⁻ cloud. Its LDFs are stronger than HCl's dipole-dipole attractions, making its boiling point higher.

Let's Practice!

1. Rank the following in order of increasing LDFs: CH₃CH₃, CH₄, CH₃CH₂CH₃



2. Rank the following in order of increasing LDFs: Br₂, F₂, Cl₂, I₂

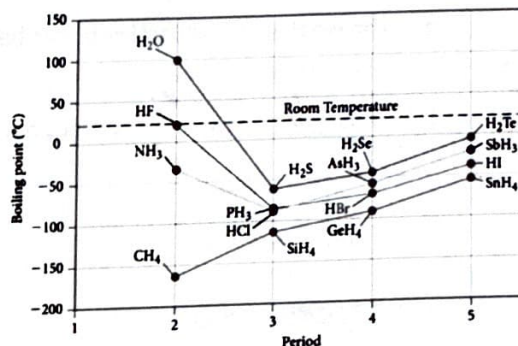


A Closer Look at Hydrogen Bonding Attractive Forces

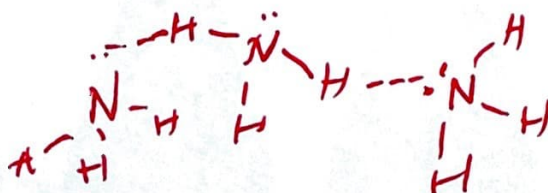
Note: Hydrogen "bonds" are NOT actual bonds (intramolecular forces), and thus the name is very misleading!

Hydrogen "Bonding": force of attraction between hydrogen atom bonded to a small highly electronegative atom (N, O, and F) and the unshared electron pair on another electronegative atom of N, O, or F.

- Hydrogen "bonding" happens with atoms that are electronegative eN-O-F! ☺
- Hydrogen "bonding" is usually depicted with a dotted or dashed line.
- Hydrogen "bonding" is responsible for some of the unique properties of water, including its relatively high boiling point.

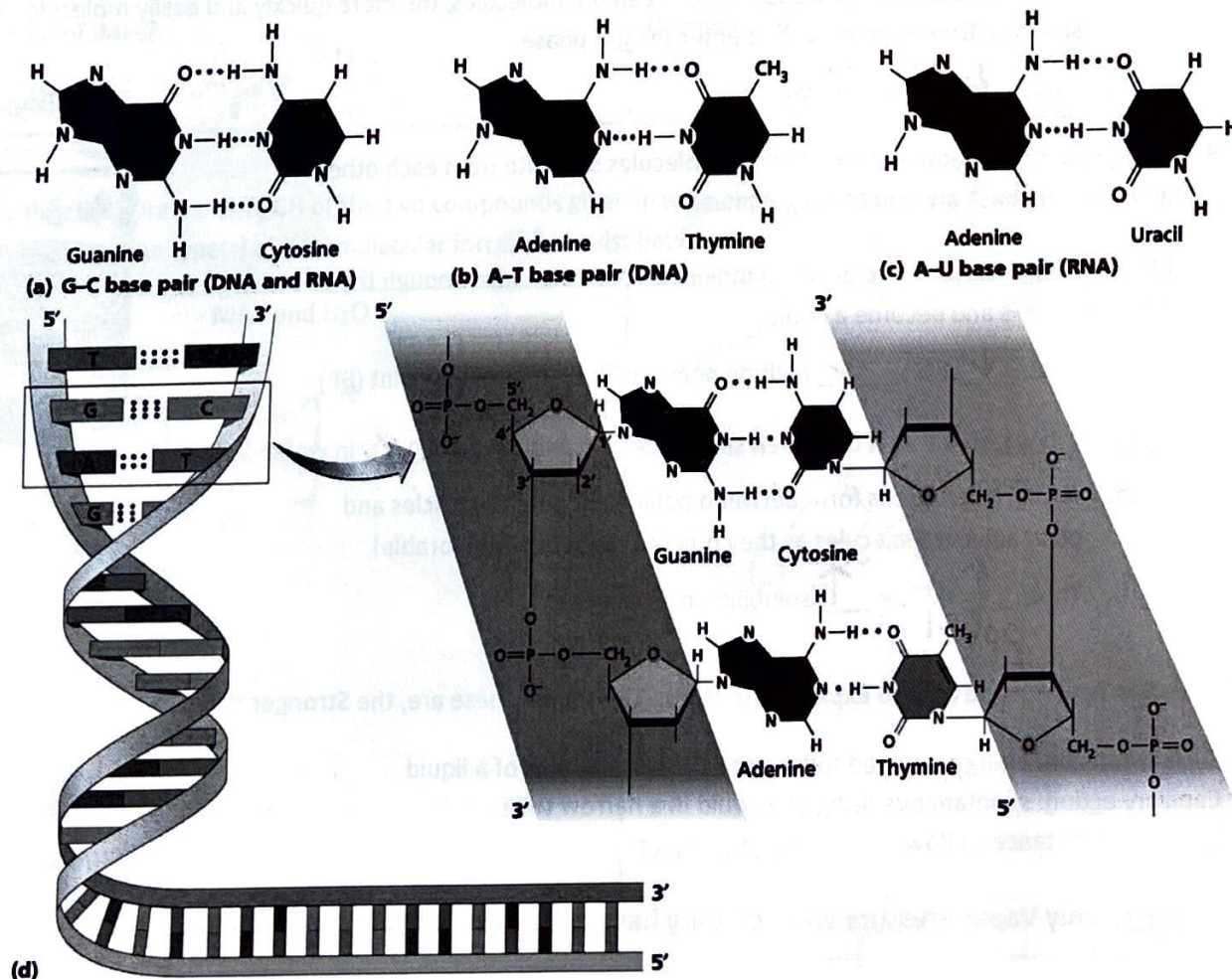


How do you represent Hydrogen Bonding? Let's look at NH₃.



Hydrogen Bonding: It's in Your DNA!

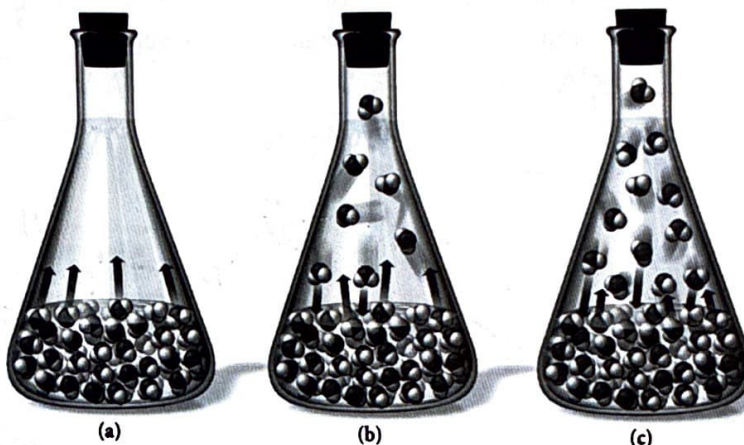
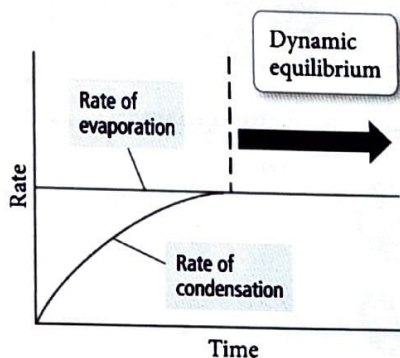
The different number of hydrogen bonds in each complementary base pair (adenine and thymine vs cytosine and guanine) helps ensure that the base pairs will match up correctly!



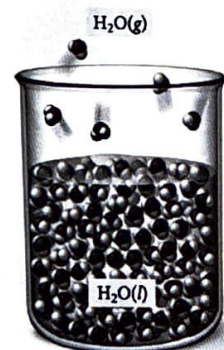
IMFs in Action

Measure of Intermolecular Forces

1. Vapor Pressure: the pressure exerted by a gas (vapor) when it is in dynamic equilibrium with its liquid (must be in closed container!)
 - a. The weaker the attractive forces between the molecules, the more molecules will be in vapor (and vice versa).
 - b. Thus, $\downarrow$ IMFs = $\uparrow$ vapor pressure (VP)



2. Volatility: how quickly a substance evaporates.
 - a. The weaker the attractive forces between the molecules, the more quickly and easily molecules will separate from each other and enter the gas phase.
 - b. Thus, $\downarrow$ IMFs = $\uparrow$ volatility
3. Boiling point: the temperature at which molecules separate from each other in the liquid phase and enter the gas phase.
4. Melting point: temperature at which molecules of a solid have enough thermal energy to overcome IMFs and become a liquid
 - a. Thus, $\downarrow$ IMFs = $\downarrow$ melting point (MP) = $\downarrow$ boiling point (BP)
5. Solubility in water: amount of a given substance that will dissolve in water
 - a. Strong interactions form between polar/ionic solute particles and polar solvent molecules as they mix $\rightarrow$ energetically favorable!
 - b. Thus, $\uparrow$ IMFs = $\uparrow$ solubility in water polar!

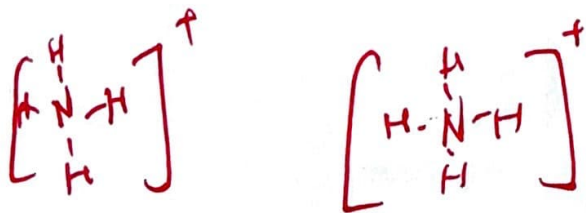


Other Properties You Must be able to Explain with IMFs: The **Higher** these are, the **Stronger** the IMFs!

1. Surface tension: energy required to increase the surface area of a liquid
2. Capillary action: spontaneous rising of a liquid in a narrow tube
3. Viscosity: resistance to flow

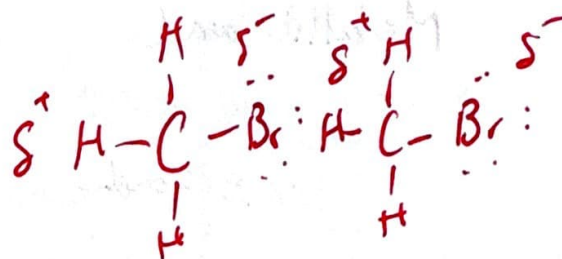
***Note:** Only Vapor Pressure and Volatility have an Inverse Relationship with IMF strength!

Let's Practice! Draw TWO Lewis dot structures for each of the following molecules. Then, draw dipoles where appropriate and determine the type(s) of intermolecular forces that exist between the molecules.



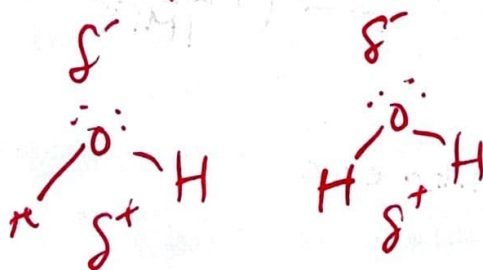
Type(s) of IMFs?

Repulsion! 2 positives



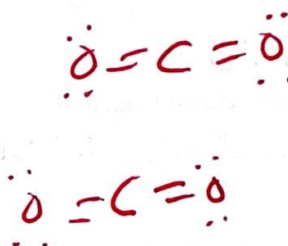
Type(s) of IMFs?

dipole-dipole, LDF



Type(s) of IMFs?

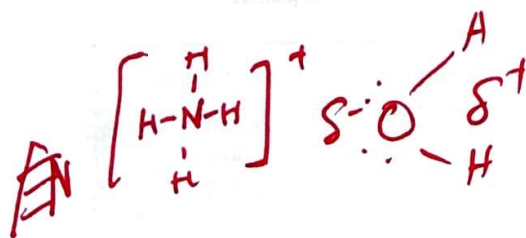
LDF, dipole-dipole H bond



Type(s) of IMFs?

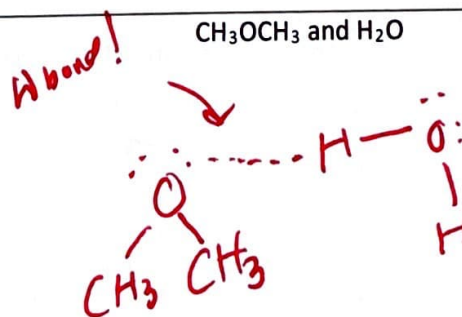
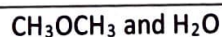
LDF

More Practice: Draw one EACH of the two compounds given in each example. Then, draw dipoles where appropriate and determine the type(s) of intermolecular forces that exist between the molecules.



Type(s) of IMFs?

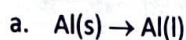
LDF, ion dipole
Strongest!



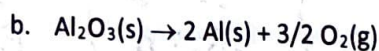
Type(s) of IMFs?

LDF dipole-dipole H bond

1. What type of intermolecular OR intramolecular attractive forces must be overcome for each process below?



Metallic bonds



ionic bonds

2. What is the predominant intermolecular force in CBr_4 ?

a. London dispersion forces

c. Dipole-dipole attraction

e. Ion-dipole attraction

b. Ionic bonding

d. Hydrogen bonding

3. Which one of the following derivatives of methane has the highest boiling point?

a. CBr_4

b. CCl_4

c. CF_4

d. CH_4

most e⁻

(~~DEF~~ IMF)

4. Based on the molecular mass and dipole moment of the five compounds in the table below, which should have the lowest vapor pressure? *← strongest IMFs = little evap*

Substance

Molecular Mass (amu)

Dipole Moment (D)

Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$

44

0.1

Dimethylether, CH_3OCH_3

46

1.3

Methylchloride, CH_3Cl

50

1.9

Acetaldehyde, CH_3CHO

44

2.7

Acetonitrile, CH_3CN

41

3.9

← most polar!

a. $\text{CH}_3\text{CH}_2\text{CH}_3$

b. CH_3OCH_3

c. CH_3Cl

d. CH_3CHO

e. CH_3CN

5. Which of the following BEST explains why neopentane has the lowest boiling point?

Common Name	n-pentane	isopentane	neopentane
Structure			
Formula	C_5H_{12}	C_5H_{12}	C_5H_{12}
Boiling Point $^{\circ}\text{C}$	36.0	27.7	9.5

a. Neopentane is less polarizable due to having fewer electrons.

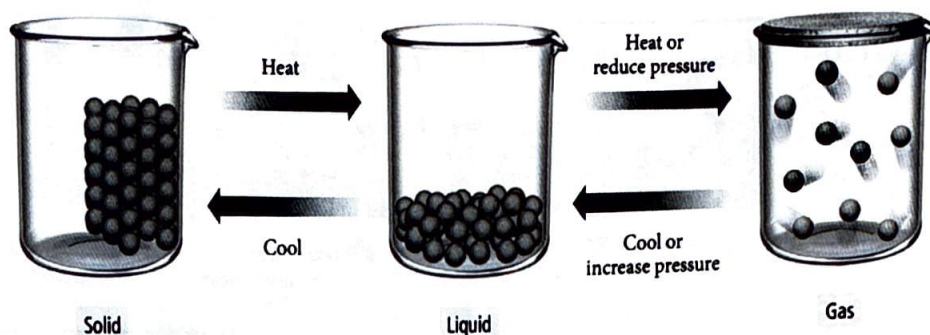
b. Neopentane is more polarizable due to having more electrons.

c. Neopentane has the shortest carbon chains and thus the least surface area.

d. Neopentane has the shortest carbon chains and thus the most surface area.

IMFs in Action = Heating Curves

KMT: The kinetic-molecular theory is based on the idea that particles of matter are always in motion.



KMT, IMFs and Changes of State: because attractive forces between the molecules are fixed, changing a material's state of matter require changing the amount of kinetic energy the particles have, or limiting their freedom.

1. **Gaseous state:** particles have complete freedom of motion.
 - a. Their kinetic energy overcomes the attractive forces between the molecules.
2. **Liquid state:** particles have limited freedom; they can move around a little within the liquid.
 - a. They have enough kinetic energy to overcome most of the attractive forces, but not enough to separate each other.
3. **Solid state:** particles are locked in place, they do not move around.
 - a. Although the particles rotate vibrate they do not have enough kinetic energy to overcome the attractive forces.

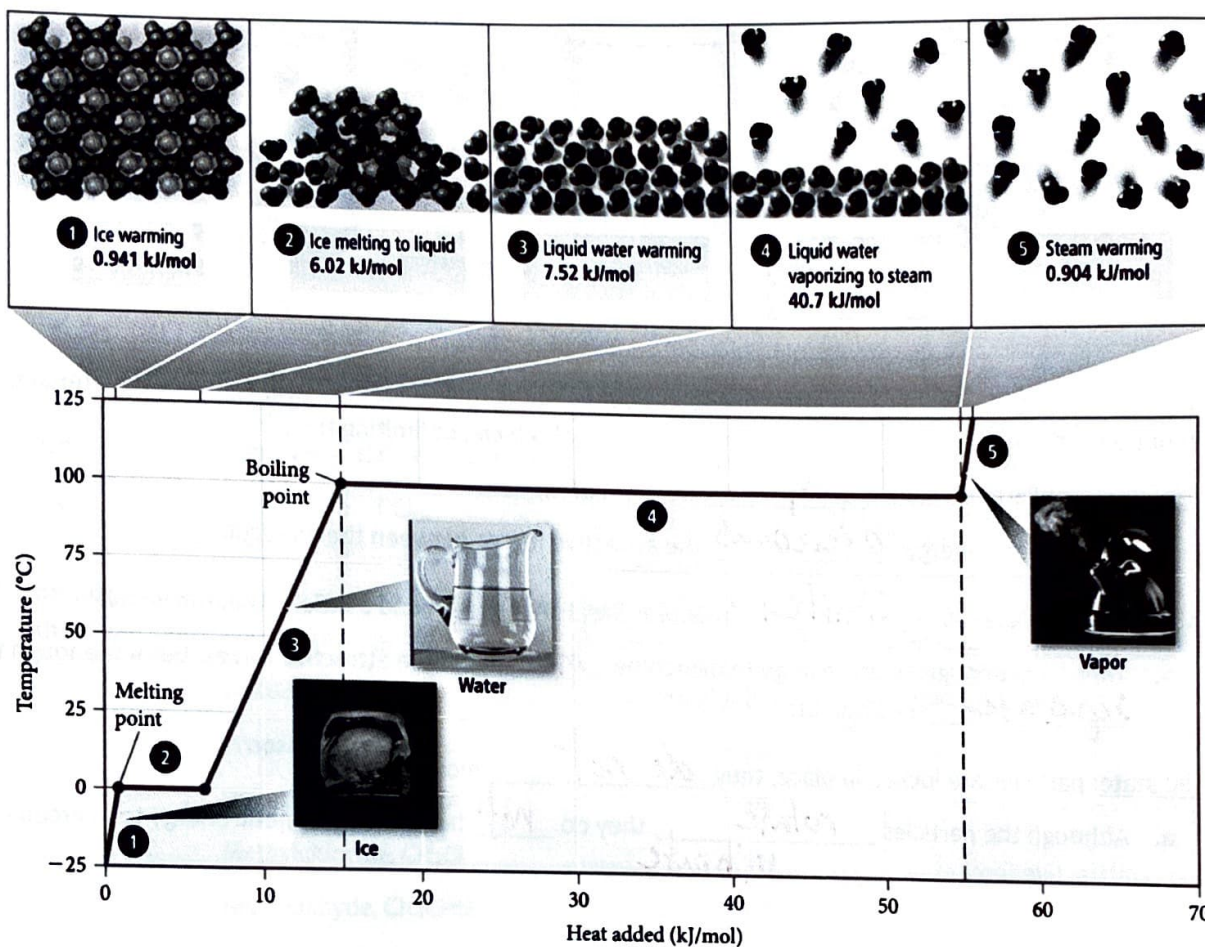
The strength of the attractive forces between particles of a substance determine its state!

- At room temperature, moderate to strong attractive forces result in materials that are solids or liquids.
- The stronger the attractive forces, the higher the boiling/ melting point!

State	Density	Shape	Volume	Strength of Intermolecular Forces (Relative to Thermal Energy)
Gas	Low	Indefinite	Indefinite	Weak
Liquid	High	Indefinite	Definite	Moderate
Solid	High	Definite	Definite	Strong

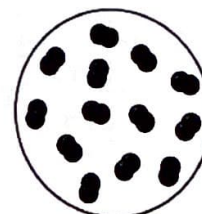
Heating and Cooling Curves

A graph of the temperature of the system versus the amount of heat added.



- In thermochemistry, our focus with heating and cooling curves is on how much thermal energy is required to change from one state of matter to another.
- For this unit, our focus is on two things:
 - the relative amount of kinetic energy for each state of matter and
 - the strength of the intermolecular forces (IMFs) holding the particles together in that state

Let's Practice! If liquid nitrogen is shown in the image to the right, which of the images below best depicts nitrogen after it has boiled?



- a) b) c) d)

Solubility and IMFs

Remember solubility rules? ☺ The bolded 3 at the beginning are the only ones you need to memorize, but there are lots and lots of solubility patterns we can observe.

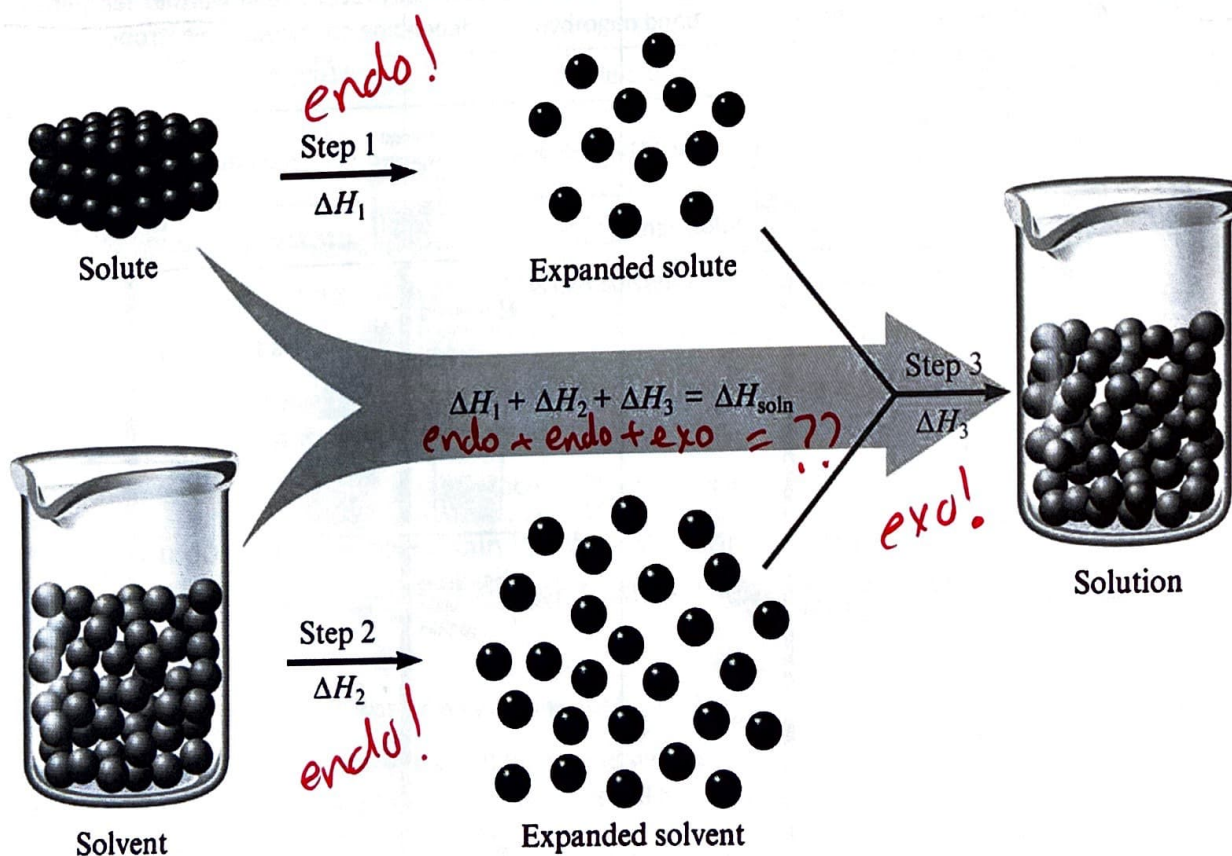
- Always soluble:** alkali metal cations, NH_4^+ , NO_3^- (also ClO_3^- , ClO_4^- , $\text{C}_2\text{H}_3\text{O}_2^-$, HCO_3^-)
- Generally soluble:**
 - Bromide, chloride, and iodide anions are soluble except when combined with Ag^+ , Pb^{2+} , and Hg_2^{2+} .
 - SO_4^{2-} is soluble except when combined with Sr^{2+} , Ba^{2+} , Pb^{2+} , and Hg_2^{2+} .
- Generally insoluble:**
 - OH^- and S^{2-} are insoluble except when combined with Ca^{2+} , Sr^{2+} , Ba^{2+} , (and things from rule 1).
 - CO_3^{2-} , PO_4^{3-} , SO_3^{2-} , and CrO_4^{2-} are insoluble except when combined with things from rule 1.

IMFs help explain these patterns of solubility!

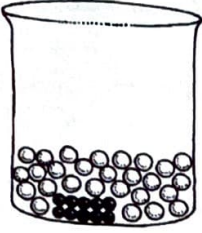
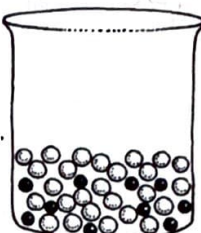
Dissolution depends on the forces of attraction between solute and solvent particles

In order to dissolve a substance, you must:

- Add energy:** Overcome attractions (requires energy = endothermic) "endo-ing" an attraction is endothermic!
 - Solute-solute IMFs (or ion-ion electrostatic attraction, if ionic)
 - Solvent-solvent IMFs
- Release energy:** Form solute-solvent attractive forces upon mixing (releases energy = exothermic)

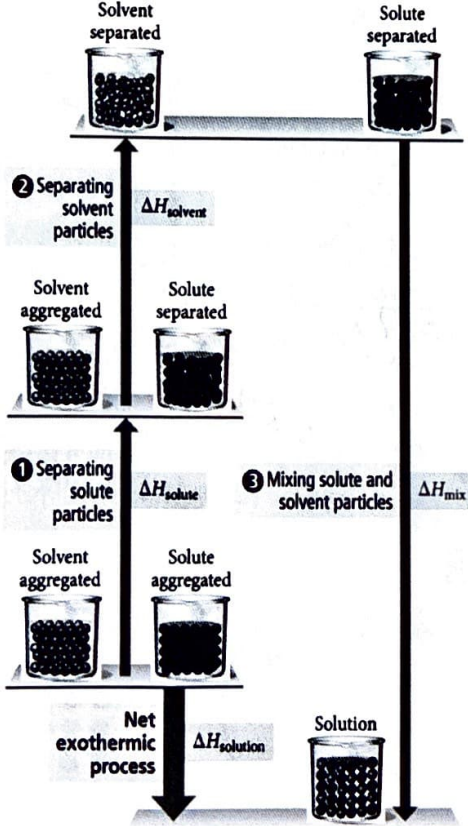
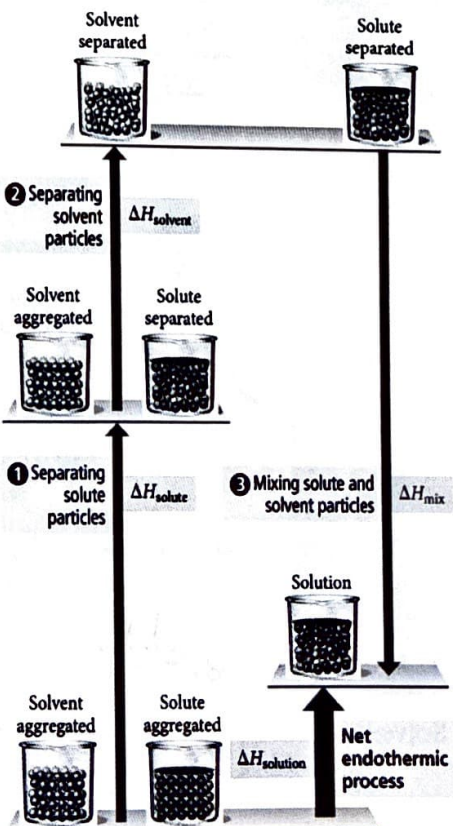


Soluble or Insoluble?

Soluble	Insoluble
Higher solute-solvent attractions	Lower solute-solvent attractions
	
Soluble (miscible)	Insoluble (immiscible)

Notice, in general, the solute particles that are **INSOLUBLE** have larger ion charges, which means they have a greater attraction to other solute particles: go, Coulomb's Law, go!

The Thermodynamics of Dissolution

Exothermic Dissolution ($-\Delta H_{\text{soln}}$)	Endothermic Dissolution ($+\Delta H_{\text{soln}}$)
- Heat released when salt dissolve - Feels warm to the touch	- Heat absorbed when salt dissolve - Feels cold to the touch
Always thermodynamically favorable ($-\Delta G$) because entropy will always increase $-\Delta H, +\Delta S$	Thermodynamically favorable at warmer temperatures, depending on increase in entropy $+\Delta H, +\Delta S$
	

Handy rule of thumb: "Like dissolves like"

Great multiple choice trick but does NOT count as explanation on free response!

1. Polar solvents dissolve polar solutes

- Hydrophilic (polar) groups to watch for: OH, CHO, C=O, COOH, NH₂, and Cl.

2. Non-polar solvents dissolve non-polar solutes

- Hydrophobic (non-polar) groups to watch for: C-H and C-C.

Many molecules have both hydrophilic and hydrophobic parts; solubility in water becomes a balance between the attraction of the polar groups for water and the attraction of the nonpolar groups for their own kind.

Never use "like dissolves like" to explain a FR on the AP exam: instead, **EXPLAIN** in terms of structure, IMFs, and energy!

So... how do you explain solubility for free response questions?

1. Identify solute-solvent IMFS

Type of Substance	Dominant Interaction with Water	Dominant Interaction with a Non-polar Solvent
Ionic	ion-dipole	ion-induced dipole (nope)
Polar + FON	hydrogen bonds	dipole-induced dipole
Polar	dipole-dipole	dipole-induced dipole
Non-polar	dipole-induced dipole	induced dipole-induced dipole (London dispersion forces)

2. Are solute-solvent attractions stronger than solute-solute (or solvent-solvent) attractions?

- Explain: strong interactions BETWEEN solvent and solute → yes, solute will dissolve!

3. Solute-solvent attractions weaker than solute-solute (or solvent-solvent) attractions?

- Explain: weak solute-solvent interactions are not as strong as existing solvent-solvent (or solute-solute) attractions, thus solute will not dissolve.

Of course, you must be SPECIFIC! Identify BOTH solute and solvent by name or formula.

*Note: you do NOT have to explain WHY a given compound can form specific IMFS; it is enough to state them.

Example #1: Can CH₃OH dissolve in water? Why or why not?

Too much "CH₃OH can form hydrogen bonds with water because it has a hydrogen which is covalently bonded to an oxygen, so it will form strong IMFs with water and thus will be able to dissolve in water."

Just right "CH₃OH can form strong hydrogen bonds with water, so it will be able to dissolve."

Not enough "CH₃OH can form strong hydrogen bonds, so it will be able to dissolve in water."

Example #2: Can benzene, C_6H_6 , dissolve in water? Why or why not?

Too much " C_6H_6 is non-polar with has a dipole moment of zero, and so it can only form weak dipole-induced dipole interactions with water, which are not as strong as the hydrogen bonds that already exist between water molecules, so C_6H_6 won't dissolve in water."

Just right " C_6H_6 is non-polar and can only form weak intermolecular attractions with water, which are not as strong as the hydrogen bonds that already exist between water molecules, so C_6H_6 won't dissolve in water."

Not enough " C_6H_6 is non-polar, so it won't dissolve in a polar substance like water."

Example FR question: Which is more likely to be soluble in water, liquid methanol (CH_3OH) or liquid hexane (C_6H_{14})? Justify your answer.

CH_3OH is polar and can form dipole dipole and H bond attractions with water, while non polar hexane can only form weaker dipole induced-dipole attractions with water.

CH_3OH is more soluble in water.

Free Response Practice!

Directions: Use principles of atomic structure, bonding, and intermolecular forces to answer the following questions. Your responses must include specific information about all substances referred to in each part.

1. Ammonia, NH_3 , is very soluble in water, whereas phosphine, PH_3 , is only moderately soluble in water. Explain.

Because NH_3 can H bond with water, the attractions are very strong. PH_3 is polar and can experience dipole dipole attractions with water, but these are not as strong making PH_3 less soluble in water than NH_3 .

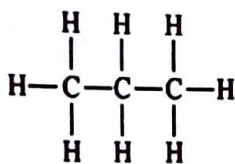
Indicate whether you agree or disagree with the statement in the box below. Justify your answer.

H_2O has a much higher boiling point than H_2S because it has very strong hydrogen bonding forces between its molecules as well as dipole-dipole forces and London dispersion forces, while H_2S has only dipole-dipole and dispersion forces. The stronger hydrogen bonding IMFs between water molecules mean that the ~~bonds~~ between water molecules are harder to ~~break~~.

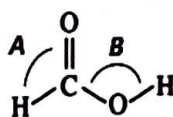
interactions/attractions

overcome

Agree w/ conclusion BUT no bonds broken! Change ~~to~~ to attractions overcome!



Propane



Methanoic Acid

3. The complete structural formulas of propane, C_3H_8 , and methanoic acid, $HCOOH$, are shown above.

a. In the table below, write the type(s) of intermolecular attractive force(s) that occur in each substance.

Substance	Boiling Point	Intermolecular Attractive Force(s)
Propane	229 K	LDF
Methanoic acid	374 K	LDF dipole-dipole H bond

b. Explain why methanoic acid has a higher boiling point than propane.

B/C methanoic acid can experience dipole-dipole & H bonds between particles, more energy must be used to overcome those attractions ~~requ~~ resulting in a higher BP. Propane's LDF attractions are weaker & easier to overcome, requiring less energy & a lower B.P.

c. Which bond angle would you predict to be larger: angle A ($H-C-O$) or angle B ($C-O-H$) in methanoic acid? Justify your answer.

B 109.5° - tetrahedral geometry

A 120° - trigonal planar geometry.